11 Mechanical Properties of Solids

Elasticity

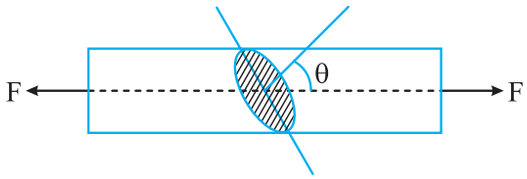
1. **Assertion (A):** The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.
Reason (R): The restoring force depends upon the bonded inter atomic and inter molecular force of solid.
(*a*) Assertion (A) is false but Reason (R) is true.
(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(*c*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(*d*) Assertion (A) is true but Reason (R) is false.

2. **Assertion (A):** Steel is used in the construction of buildings and bridges.
Reason (R): Steel is more elastic and its elastic limit is high.
(*a*) Assertion (A) is false but Reason (R) is true.
(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(*c*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(*d*) Assertion (A) is true but Reason (R) is false.

Stress

3. The ratio of radii of two wires of same material is 2 : 1. If these wires are stretched by equal force, the ratio of stresses produced in them is:
(*a*) 2 : 1 (*b*) 1 : 2 (*c*) 1 : 4 (*d*) 4 : 1

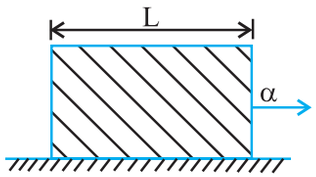
4. A bar of cross section A is subjected to equal and opposite tensile forces at the ends as shown below:



Let the normal to the cross section of the bar makes an angle θ with the axis of the bar. The tensile stress on this plane is

(*a*) $\frac{F\cos^2\theta}{A}$ (*b*) $\frac{F}{A\cos^2\theta}$ (*c*) $\frac{F\cos\theta}{A}$ (*d*) $\frac{F}{A\cos\theta}$

5. A uniform rod of length L and density ρ is being pulled along a smooth floor with a horizontal acceleration α (see figure). The magnitude of the stress at the transverse cross-section through the mid-point of the rod is .



(*a*) $\frac{1}{2}\rho\alpha L$ (*b*) $\frac{3}{2}\rho\alpha L$ (*c*) $\frac{5}{2}\rho\alpha L$ (*d*) $2\rho\alpha L$

Strain

6. A uniform cubical block is subjected to volumetric compression, which decreases its each side by 2%. The Bulk strain produced in it is:
(*a*) 0.03 (*b*) 0.02 (*c*) 0.06 (*d*) 0.12

7. Correct pair is:
(*a*) Change in shape - Longitudinal strain
(*b*) Change in volume - Shear strain
(*c*) Change in length - Bulk strain
(*d*) Reciprocal of Bulk modulus - Compressibility

8. Match the following type questions.

Column-I		Column-II	
A.	Longitudinal stress	p.	Volume changes
B.	Shear stress	q.	Shape changes
C.	Volumetric stress	r.	Volume does not change
D.	Tensile stress	s.	Shape does not change only size changes

(*a*) A-(p,s), B-(q,r), C-(p,s), D-(p,s)
(*b*) A-(q,r), B-(p,s), C-(p,s), D-(p,s)
(*c*) A-(p,s), B-(p,s), C-(q,r), D-(p,s)
(*d*) A-(p,s), B-(p,s), C-(p,s), D-(q,r)

9. **Statement-I:** Strain is a unit less quantity.
Statement-II: Strain is equivalent to force.
Choose the correct option
(*a*) Statement-I and Statement-II both are correct.
(*b*) Statement-I and Statement-II both are incorrect.
(*c*) Statement-I is correct and Statement-II is incorrect.
(*d*) Statement-I is incorrect but Statement-II is correct.

10. Four wires P,Q,R and S of same materials have diameters and stretching forces as shown below. Arrange their strains in the decreasing order.

Wire	Diameter	Stretching force
P	2 mm	10 N
Q	1 mm	20 N
R	4 mm	30 N
S	3 mm	40 N

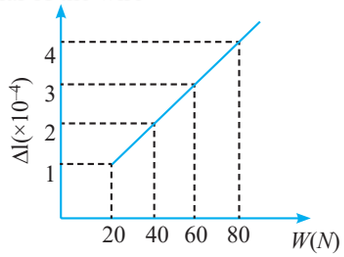
(*a*) Q,S,P,R (*b*) R,P,S,Q (*c*) P,Q,R,S (*d*) P,R,Q,S

Young's Modulus

11. With rise in temperature, the Young's modulus of elasticity:
(*a*) May increase or decrease (*b*) Increases
(*c*) Decreases (*d*) Remains unchanged

12. Young's modulus of material of wire of length 'L' and cross-sectional area A is Y. If the length of the wire is doubled and radius is decreased by 1/4th of it's initial radius then Young's modulus will be:
(*a*) $\frac{32}{9}Y$ (*b*) $8Y$ (*c*) $\frac{8}{3}Y$ (*d*) Y

13. The adjacent graph shows the extension (Δl) of a wire of length 1*m* suspended from the top of a roof at one end with a load *W* connected to the other end. If the cross sectional area of the wire is $10^{-6}m^2$ calculate the young's modulus of the material of the wire

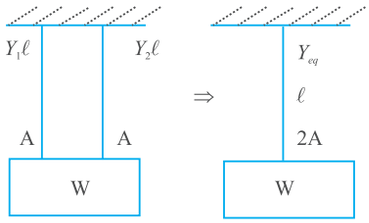


(*a*) $2 \times 10^{11}\ N/m^2$ (*b*) $2 \times 10^{-11}\ N/m^2$
(*c*) $3 \times 10^{-12}\ N/m^2$ (*d*) $2 \times 10^{-13}\ N/m^2$

14. The temperature of a wire is doubled. The Young's modulus of elasticity:
(*a*) Will also double (*b*) Will become four times
(*c*) Will remain same (*d*) Will decrease

15. Modulus of rigidity of ideal liquids is:
(*a*) Infinity
(*b*) Zero
(*c*) Unity
(*d*) Some finite small non-zero constant value

16. Two wires of same length and radius are joined in parallel and loaded. The Young's modulii of the material of the wires are Y_1 & Y_2. If the combination is taken as a single wire then its Young's modulus is



(*a*) $\frac{Y_1Y_2}{Y_1+Y_2}$ (*b*) $\frac{Y_1+Y_2}{2}$
(*c*) $\frac{2Y_1Y_2}{Y_1+Y_2}$ (*d*) $\frac{Y_1+Y_2}{Y_1Y_2}$

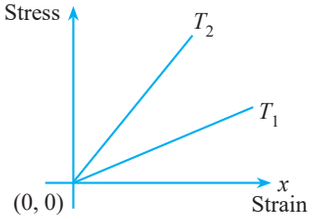
17. The Young's modulus of brass and steel are respectively $1.0 \times 10^{11}\ Nm^{-2}$ and $2.0 \times 10^{11}\ Nm^{-2}$. A brass wire and a steel wire of the same length are extended by 1 mm each under the same force. If radii of brass and steel wires are R_B and R_S respectively, then
(*a*) $R_S = \sqrt{2}R_B$ (*b*) $R_s = \frac{R_B}{\sqrt{2}}$
(*c*) $R_S = 4R_B$ (*d*) $R_s = \frac{R_B}{2}$

18. **Statement-I:** Identical springs of steel and copper are equally stretched. More work will be done on the steel spring.
Statement-II: Steel is more elastic than copper.
(*a*) Statement-I and Statement-II both are correct.
(*b*) Statement-I and Statement-II both are incorrect.
(*c*) Statement-I is correct and Statement-II is incorrect.
(*d*) Statement-I is incorrect but Statement-II is correct.

19. A rigid bar of mass M is supported symmetrically by three wires each of length *l*. Those at each end are of copper and the middle one is of iron. The ratio of their diameters, if each having the same tension, is equal to
(*a*) $\sqrt{\frac{Y_{copper}}{Y_{iron}}}$ (*b*) $\sqrt{\frac{Y_{iron}}{Y_{copper}}}$ (*c*) $\frac{Y^2_{iron}}{Y^2_{copper}}$ (*d*) $\frac{Y_{iron}}{Y_{copper}}$

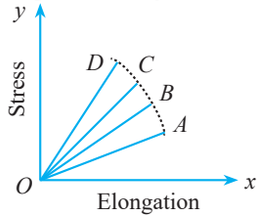
Hook's Law

20. Figure shows graph between stress and strain for a uniform wire at two different temperatures. Then,



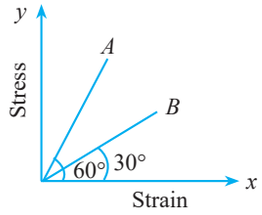
(*a*) $T_1 > T_2$ (*b*) $T_2 > T_1$
(*c*) $T_1 = T_2$ (*d*) None of these

21. The load versus elongation graph for four wires of same length and same material is shown in figure. The thinnest wire is represented by line.



(*a*) OC (*b*) OD (*c*) OA (*d*) OB

22. The stress versus strain graph for wires of two materials *A* and *B* are as shown in the figure. If Y_A and Y_B are the Young's moduli of the materials, then

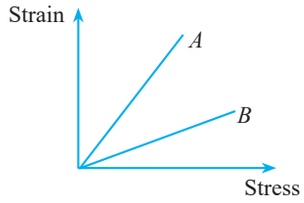


(*a*) $Y_B = 2Y_A$ (*b*) $Y_A = 3Y_B$

(*c*) $Y_B = 3Y_A$ (*d*) $Y_A = Y_B$

23. For the given graph, answer in True and False.

A wire elongates more for same force applied to A and B having same length and area of cross-section.



24. Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3A. If the length of the first wire is increased by Δ1 on applying a force F, how much force is needed to stretch the second wire by the same amount?

(*a*) 9*F* (*b*) 6*F* (*c*) *F* (*d*) 4*F*

25. The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross-section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to the steel and brass wires must be in the ratio of:

(*a*) 1 : 1 (*b*) 1 : 2 (*c*) 2 : 1 (*d*) 4 : 1

26. The following four wires of length L, and radius *r* are made of the same material. Which of these will have the largest extension, when the same tension is applied?

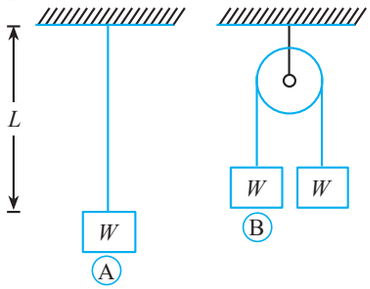
(*a*) $L = 400$ cm, $r = 0.8$ mm (*b*) $L = 300$ cm, $r = 0.6$ mm

(*c*) $L = 200$ cm, $r = 0.4$ mm (*d*) $L = 100$ cm, $r = 0.2$ mm

27. Two wire *A* and *B* are stretched by same force. If, for *A* and *B*, $Y_A : Y_B = 1 : 2$, $r_A : r_B = 3 : 1$ and $l_A : l_B = 4 : 1$, then ratio of their extension $\left(\frac{\Delta l_A}{\Delta l_B}\right)$ will be:

(*a*) 10 : 13 (*b*) 11 : 7 (*c*) 8 : 9 (*d*) 6 : 5

28. If in case *A*, elongation in wire of length *L* is *l*, then for same wire elongation in case *B* will be:



(*a*) 4*l* (*b*) 2*l* (*c*) *l* (*d*) *l*/2

29. A wire of length *L* and radius *r* fixed at one end and a force *F* applied to the other end produces an extension *l*. The extension produced in another wire of the same material of length 2*L* and radius 2*r* by a force 2*F*, is:

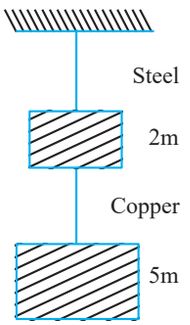
(*a*) *l* (*b*) 2*l* (*c*) 4*l* (*d*) *l*/2

30. A wire of length *L*, area of cross section *A* is hanging from a fixed support. The length of the wire changes to L_1 when mass *M* is suspended from its free end. The expression for Young's modulus is:

(*a*) MgL_1/AL (*b*) $Mg(L_1 - L)/AL$

(*c*) MgL/AL_1 (*d*) $MgL/A(L_1 - L)$

31. If the ratio of diameters, lengths and Young's modulus of steel and copper wires shown in the figure are *p*, *q* and *s* respectively, then the corresponding ratio of increase in their lengths would be:

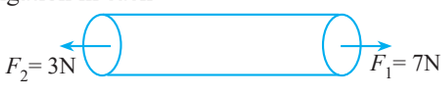


(*a*) $5q/(7sp^2)$ (*b*) $7q/(5sp^2)$

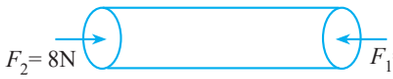
(*c*) $2q/(5sp)$ (*d*) $7q/(5sp)$

32. If length, cross section area and young moduless of the following figures be l, A and Y respectively then find the elongation in each

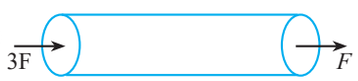
(*i*)



(*ii*)



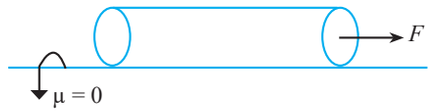
(*iii*)



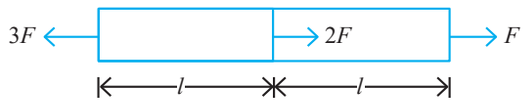
(iv)



(v)



33. A bar is subjected to axial forces as shown. If E is the modulus of elasticity of the bar and A is its cross-section area. Its elongation will be:



(a) $\frac{Fl}{AE}$ (b) $\frac{2Fl}{AE}$ (c) $\frac{3Fl}{AE}$ (d) $\frac{4Fl}{AE}$

34. A wire of length L and radius r is clamped at one end. If it's other end is pulled by a force F, its length increases by l. If the radius of wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become.

(a) 2 times (b) 3 times (c) 4 times (d) 3/2 times

35. The Young's modulus of a steel wire of length 6 m and cross-sectional area 3 mm^2, is 2×10^{11} N/m^2. The wire is suspended from its support on a given planet. A block of mass 4 kg is attached to the free end of the wire. The acceleration due to gravity on the planet is $1/4^{th}$ of its value on the earth. The elongation of wire is (Take g on the earth = 10 m/s^2):

(a) 0.1 cm (b) 1 cm

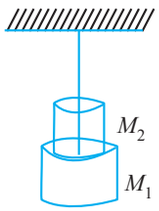
(c) 0.1 mm (d) 1 mm

36. What is the percentage increase in length of a wire of diameter 2.5 mm, stretched by a force of 100 kg wt? Young's modulus of elasticity of wire = 12.5×10^{11} dyne/cm^2.

(a) 0.16% (b) 0.32%

(c) 0.08% (d) 0.12%

37. The length of wire, when M_1 is hung from it, is l_1 and is l_2 with both M_1 and M_2 hanging. The natural length of wire is:



(a) $\frac{M_1}{M_2}(l_1 - l_2) + l_1$ (b) $\frac{M_2 l_1 - M_1 l_2}{M_1 + M_2}$

(c) $\frac{l_1 + l_2}{2}$ (d) $\sqrt{l_1 l_2}$

38. The lengths of a metallic wire are L_1 and L_2 when the tensions in the wire are T_1 and T_2 respectively. Find the natural length of the wire.

39. The length of an elastic string is X m when the tension is $8N$, and Y m when the tension is 10 N. The length in metres when the tension is 18 N is:

(a) $4X - 5Y$ (b) $5Y - 4X$

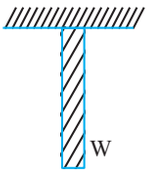
(c) $9X - 4Y$ (d) $4Y - 9X$

40. A wire of length L is hanging from a fixed support. The length changes to L_1 and L_2 when masses 1kg and 2kg are suspended respectively from its free end. Then the value of L is equal to:

(a) $\sqrt{L_1 + L_2}$ (b) $\frac{L_1 + L_2}{2}$

(c) $2L_1 - L_2$ (d) $3L_1 - L_2$

41. A rod of uniform cross sectional area A and length L has a weight W. It is suspended vertically from a fixed support. If Young's modulus for rod is Y, then elongation produced in rod is:



(a) $\frac{WL}{YA}$ (b) $\frac{WL}{2YA}$ (c) $\frac{WL}{4YA}$ (d) $\frac{3WL}{4YA}$

42. The bulk modulus of a spherical object is B. If it is subjected to uniform pressure p, the frictional decrease in radius is:

(a) $\frac{p}{B}$ (b) $\frac{B}{3p}$

(c) $\frac{3p}{B}$ (d) $\frac{p}{3B}$

43. A uniform cylindrical rod of length L and radius r, is made from a material whose Young's modulus of Elasticity equals Y. When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F, its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is (nearly) equal to:

(a) $\frac{3F}{(\pi r^2 YT)}$ (b) $\frac{6F}{(\pi r^2 YT)}$

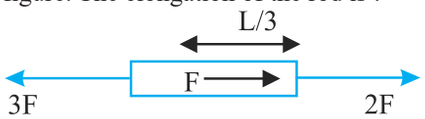
(c) $\frac{F}{(3\pi r^2 YT)}$ (d) $9F(\pi r^2 YT)$

44. Copper of fixed volume V is drawn into wire of length l. When this wire is subjected to a constant force F, the extension produced in the wire is Δl. Which of the following graphs is a straight line?

(a) Δl versus $\frac{1}{l}$ (b) Δl versus l^2

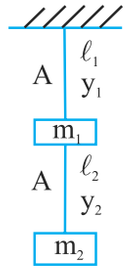
(c) Δl versus $\frac{1}{l^2}$ (d) Δl versus l

45. A uniform slender rod of length L, cross- sectional area A and Young's modulus Y is acted upon by the forces shown in the figure. The elongation of the rod is :



(*a*) $\frac{3FL}{5AY}$ (*b*) $\frac{2FL}{5AY}$ (*c*) $\frac{3FL}{8AY}$ (*d*) $\frac{8FL}{3AY}$

46. Two wires are arranged as shown in the figure. The elongations of upper and lower wires are respectively



(*a*) $\frac{(m_1+m_2)g\ell_1}{Ay_1}, \frac{m_2g\ell_2}{Ay_2}$ (*b*) $\frac{(m_1-m_2)g\ell_1}{Ay_1}, \frac{m_2g\ell_2}{Ay_2}$

(*c*) $\frac{\left(\frac{m_1}{m_2}+1\right)g\ell_1}{Ay_1}, \frac{m_2g\ell_2}{Ay_2}$ (*d*) $\frac{\left(\frac{m_1}{m_2}-1\right)g\ell_1}{Ay_1}, \frac{m_2g\ell_2}{Ay_2}$

47. Two exactly similar wires of steel ($Y = 20 \times 10^{11}$ dyne/cm^2) and copper ($Y = 12 \times 10^{11}$ dyne/cm^2) are stretched by equal forces. If the total elongation is 1cm, elongation of copper wire is

(*a*) 3/5 cm (*b*) 5/3 cm (*c*) 3/8 cm (*d*) 5/8 cm

Bulk Modulus

48. A spherical ball contracts in volume by 0.01% when subjected to a normal uniform pressure of 100 atm. The Bulk modulus of its material is:

(*a*) 1.01×10^{11} N m^{-2} (*b*) 1.01×10^{12} N m^{-2}

(*c*) 1.01×10^{10} N m^{-2} (*d*) 1.01×10^{13} N m^{-2}

49. If a rubber ball is taken at the depth of 200 m in a pool, its volume decreases by 0.1%. If the density of the water is 1×10^3 kg/m^3 and $g = 10$ m/s^2, then the volume elasticity in will be:

(*a*) 10^8 (*b*) 2×10^8 (*c*) 10^9 (*d*) 2×10^9

50. Select the incorrect statement about Bulk modulus of elasticity.

(*a*) It is defined for solids, liquids and gases.

(*b*) $B_{solid} > B_{liquid} > B_{gas}$

(*c*) The bulk modulus of gas is different for different processes.

(*d*) Almost for all materials, the Bulk modulus increases with the rise in temperature.

51. An object is located at 2 km below the surface of the water. If the fractional compression $\Delta V/V$ is 1.36%, the ratio of hydraulic stress to the corresponding hydraulic strain will be_____.

[Given: density of water is 1000 kgm^{-3} and g = 9.8 ms^{-2}]

(*a*) 1.44×10^7 Nm^{-2} (*b*) 1.44×10^9 Nm^{-2}

(*c*) 1.96×10^7 Nm^{-2} (*d*) 2.26×10^9 Nm^{-2}

52. A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to:

(*a*) 0.6 (*b*) 20 (*c*) 1.67 (*d*) 5

53. Match the Column-I with Column-II

Column-I		Column-II	
A.	A body which regains its original shape after the removal of external force.	p.	Elasticity
B.	A body which does not regain its original shape after the removal of external force.	q.	Elastic body
C.	A body which does not show any deformation on applying external force.	r.	Plastic body
D.	The property of the body to regain its original configuration when the deforming forces are removed.	s.	Rigid body

(*a*) A-(p), B-(r), C-(s), D-(q) (*b*) A-(q), B-(s), C-(r), D-(p)

(*c*) A-(q), B-(r), C-(s), D-(p) (*d*) A-(p), B-(s), C-(q), D-(r)

54. **Statement-I:** Bulk modulus of elasticity (K) represents incompressibility of the material.

Statement-II: Bulk modulus of elasticity is proportional to change in pressure.

(*a*) Statement-I and Statement-II both are correct.

(*b*) Statement-I and Statement-II both are incorrect.

(*c*) Statement-I is correct and Statement-II is incorrect.

(*d*) Statement-I is incorrect but Statement-II is correct.

55. A cubical ball is taken to a depth of 200m in a sea. The decrease in volume observed to be 0.1% the bulk modulus of the ball is ($g = 10$ m/s^2)

(*a*) 2×10^7 *pa* (*b*) 2×10^6 *pa*

(*c*) 2×10^9 pa (*d*) 1.2×10^9 *pa*

56. What will be the percentage fractional compression of water at the bottom of the ocean about 5000 m deep given that the bulk modulus of water is 3×10^9 Pa. $g = 9.8$ m/s^2 (1Pa = 1 N/m^2)

(*a*) 6.8% (*b*) 1.6% (*c*) 2.4% (*d*) 18%

Bulk Modulus of a Gas in Different Processes

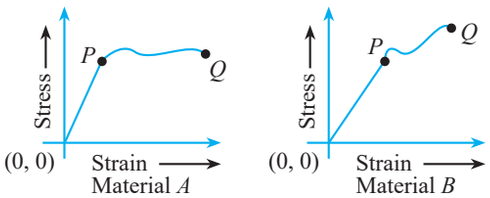
57. Match the following column-I and column-II

Column-I		Column-II	
A.	Isobaric process	I.	$\beta = \gamma P$
B.	Isochoric Process	II.	$\beta = 0$ (zero)
C.	Adiabatic Process	III.	$\beta = \infty$ (Infinite)
D.	Isothermal Process	IV.	$\beta = \Pi$ (Pressure)

(*a*) A-IV, B-I, C-II, D-III
(*b*) A-I, B-II, C-III, D-IV
(*c*) A-III, B-IV, C-I, D-II
(*d*) A-II, B-III, C-I, D-IV

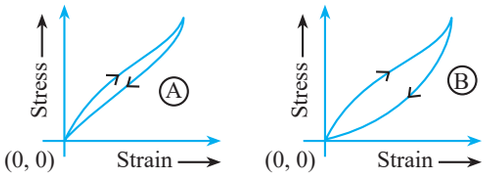
Stress-Strain Graph

58. The stress-strain graphs for two materials *A* and *B* are shown in figure. The graphs are drawn to the same scale. Select the correct statement.



(*a*) Material *A* has greater Young's Modulus
(*b*) Material *A* is ductile
(*c*) Material *B* is brittle
(*d*) All of these

59. Two different types of rubber are found to have the stress-strain curves as shown. Then-

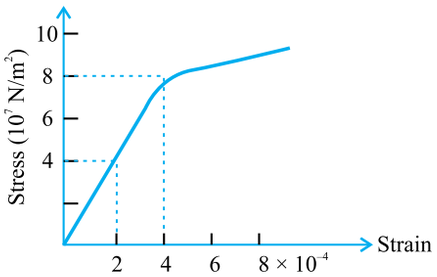


(*a*) *A* is suitable for shock absorber
(*b*) *B* is suitable for shock absorber
(*c*) *B* is suitable for car tyres
(*d*) None of these

60. The stress-strain curves are down for two different materials *X* and *Y*. It is observed that the ultimate strength point and the fracture point are close to each other for material *X* but are far apart for material *Y*. We can say that materials *X* and Y are likely to be (respectively).

(*a*) Plastic and ductile (*b*) Ductile and brittle
(*c*) Brittle and ductile (*d*) Brittle and plastic

61. Which one of the following is the Young's modulus (in N/m^2) for the wire having the stress-strain curve shown in the figure



(*a*) 9×10^4 (*b*) 8.0×10^{11}
(*c*) 10×10^{11} (*d*) 2.0×10^5

Breaking Force & Breaking Stress

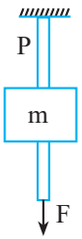
62. The maximum load a wire can withstand without breaking, when its length is reduced to half of its original length, will:

(*a*) Be half (*b*) Be double
(*c*) Be four times (*d*) Remain same

63. The maximum length of a wire having breaking stress 8×10^8 N/m^2 and density 8×10^3 kg/m^3 that can be suspended from a rigid support without breaking is ($g = 10 m/s^2$)

(*a*) 5 km (*b*) 12 km (*c*) 10 km (*d*) 24 km

64. Two light wires P and Q shown in the figure are made of same material and have radii r_P and r_Q, respectively. The block between them has a mass m. When the force $F = \frac{mg}{3}$, then one of the wires breaks. Read the following statement and choose correct answer



A. P breaks, if $r_P = r_Q$
B. P breaks, if $r_P < 2r_Q$
C. Either P or Q may breaks if $r_P = 2r_Q$
D. To predict, which wire will break, the lengths of P and Q must be known.

(*a*) Only A, B & C are correct (*b*) Only B & D are correct
(*c*) Only A & D are correct (*d*) Only A & B are correct

65. A wire of length L and density ρ and Young's modulus Y is hanging from a support. Find the elongation in the length of wire at which wire will break :

(*a*) $\frac{L^2\rho g}{Y}$ (*b*) $\frac{L^2\rho g}{2Y}$ (*c*) $\frac{2L^2\rho g}{Y}$ (*d*) $\frac{L^2\rho g}{4Y}$

66. A cable wire which can support a load of 800 N is cut into two equal parts. The maximum load that can be supported by either part of the cable wire:

(*a*) 200 N (*b*) 400 N (*c*) 800 N (*d*) 1200 N

Answer Key

1. (*c*)	**2.** (*c*)	**3.** (*c*)	**4.** (*a*)	**5.** (*a*)	**6.** (*c*)	**7.** (*d*)	**8.** (*a*)	**9.** (*c*)	**10.** (*a*)
11. (*c*)	**12.** (*d*)	**13.** (*a*)	**14.** (*d*)	**15.** (*b*)	**16.** (*b*)	**17.** (*b*)	**18.** (*a*)	**19.** (*b*)	**20.** (*a*)
21. (*c*)	**22.** (*b*)	**23.** (True)	**24.** (*a*)	**25.** (*c*)	**26.** (*d*)	**27.** (*c*)	**28.** (*a*)	**29.** (*a*)	**30.** (*d*)

31. (*b*) **32.** (*i*) $\Delta l = \frac{5l}{AY}$, (*ii*) $\Delta l = \frac{11l}{2AY}$, (*iii*) $\Delta l = \frac{Fl}{AY}$, (*iv*) $\Delta l = \frac{l}{2AY}$, (*v*) $\Delta l = \frac{Fl}{2AY}$ **33.** (*d*) **34.** (*a*) **35.** (*c*)

36. (*a*)	**37.** (*a*)	**38.** $\frac{L_1T_2 - L_2T_1}{T_2 - T_1}$		**39.** (*b*)	**40.** (*c*)	**41.** (*b*)	**42.** (*d*)	**43.** (*a*)	**44.** (*d*)
45. (*d*)	**46.** (*a*)	**47.** (*d*)	**48.** (*a*)	**49.** (*d*)	**50.** (*d*)	**51.** (*b*)	**52.** (*c*)	**53.** (*c*)	**54.** (*a*)
55. (*c*)	**56.** (*b*)	**57.** (*d*)	**58.** (*d*)	**59.** (*b*)	**60.** (*c*)	**61.** (*d*)	**62.** (*d*)	**63.** (*c*)	**64.** (*a*)
65. (*a*)	**66.** (*c*)								